

ASSIGNMENT : - 4

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Ans: -

No.	Scenario	How the Multimeter Helps
1.	Computer power supply not working	Checks the voltage output of the power supply to see if it is supplying the correct power.
2.	Laptop battery not charging	Measures the battery voltage to check whether the battery is receiving or holding the correct voltage.
3.	Smartphone not turning on	Checks the battery voltage and continuity to identify a faulty battery, charging connection, or circuit.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

Ans: -

A POST card displays hexadecimal codes as the BIOS checks different motherboard hardware during startup.

In the demonstration, the codes changed during the BIOS testing and finally displayed 8080, indicating that POST completed successfully.

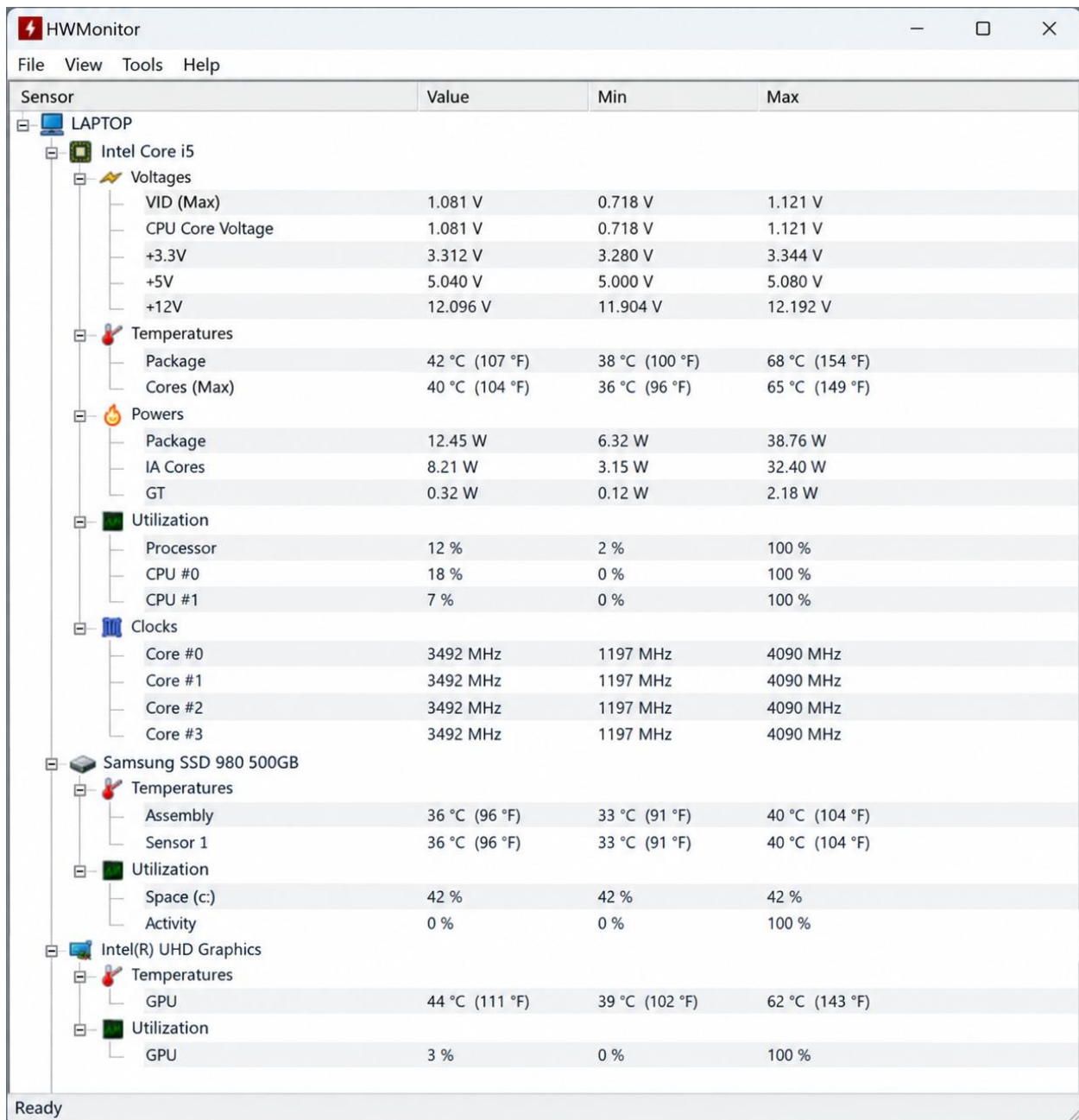
The card also has LED indicators for **power voltages, CPU/clock, reset, and other motherboard signals.

It helps identify hardware problems by showing the last POST code reached when the system stops booting, narrowing the problem to a particular hardware stage.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.

Ans: -



The screenshot shows the HWMonitor application window. The title bar is 'HWMonitor' with standard window controls. The menu bar includes 'File', 'View', 'Tools', and 'Help'. The main content area is a table with columns 'Sensor', 'Value', 'Min', and 'Max'. The sensors are organized into a tree view on the left, including 'LAPTOP', 'Intel Core i5', 'Samsung SSD 980 500GB', and 'Intel(R) UHD Graphics'. Each sensor has sub-sensors for various metrics like voltages, temperatures, powers, utilization, and clocks. The status bar at the bottom left says 'Ready'.

Sensor	Value	Min	Max
LAPTOP			
Intel Core i5			
Voltages			
VID (Max)	1.081 V	0.718 V	1.121 V
CPU Core Voltage	1.081 V	0.718 V	1.121 V
+3.3V	3.312 V	3.280 V	3.344 V
+5V	5.040 V	5.000 V	5.080 V
+12V	12.096 V	11.904 V	12.192 V
Temperatures			
Package	42 °C (107 °F)	38 °C (100 °F)	68 °C (154 °F)
Cores (Max)	40 °C (104 °F)	36 °C (96 °F)	65 °C (149 °F)
Powers			
Package	12.45 W	6.32 W	38.76 W
IA Cores	8.21 W	3.15 W	32.40 W
GT	0.32 W	0.12 W	2.18 W
Utilization			
Processor	12 %	2 %	100 %
CPU #0	18 %	0 %	100 %
CPU #1	7 %	0 %	100 %
Clocks			
Core #0	3492 MHz	1197 MHz	4090 MHz
Core #1	3492 MHz	1197 MHz	4090 MHz
Core #2	3492 MHz	1197 MHz	4090 MHz
Core #3	3492 MHz	1197 MHz	4090 MHz
Samsung SSD 980 500GB			
Temperatures			
Assembly	36 °C (96 °F)	33 °C (91 °F)	40 °C (104 °F)
Sensor 1	36 °C (96 °F)	33 °C (91 °F)	40 °C (104 °F)
Utilization			
Space (c:)	42 %	42 %	42 %
Activity	0 %	0 %	100 %
Intel(R) UHD Graphics			
Temperatures			
GPU	44 °C (111 °F)	39 °C (102 °F)	62 °C (143 °F)
Utilization			
GPU	3 %	0 %	100 %

Ready

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Ans: -

Tool	What it checks	Example Problem Detected
Multimeter	Measures voltage, resistance, and continuity in electronic components.	Dead laptop battery or faulty power connection.
POST Card	Checks motherboard startup/POST codes during boot.	RAM, CPU, or motherboard failure preventing the computer from booting.
Software Diagnostics	Monitors hardware health, temperatures, storage, and system performance.	Overheating CPU or failing/ unhealthy SSD.